

Square (Smart Parcel Box)

2802416835 – Rafael Marvin Jonathan, 2802391681 – Eric Christian Darmawan, 2802428072 –
Nicholas Laurensius Halim
LC40 / John Reigton Hartono, Genap 2025-2026

1. Introduction (Context & Motivation):

- **Relevance:**

The rise of e-commerce has escalated "last-mile" delivery issues, specifically package theft (porch piracy) and unverified deliveries. Traditional drop-boxes are passive and cannot authenticate couriers or confirm package drops. This problem strictly requires an IoT solution to transform a vulnerable container into an active, secure, and verifiable smart vault.

- **IoT Appropriateness:**

- **Edge Sensing:** An ESP32 is required to process local inputs and trigger immediate physical security responses without latency.
- **Cloud Sync:** Middleware is essential to securely store photo evidence off-site and validate dynamic courier PINs, ensuring data remains safe even if the physical box is tampered with.
- **Mobile Monitoring:** A mobile app bridges the physical distance, allowing users to actively manage security via remote Live Cam, and other features.

- **Evidence:**

The system integrates ultrasonic waves to detect box overload, electromagnetic actuation (solenoid) for access control, and mechanical efficiency, utilizing the door's hinge to mechanically shift a single camera's FOV. This solution also directly mitigates financial losses from package theft and establishes a secure delivery standard that provides peace of mind.

2. Problem Definition and Objectives:

- **Problem Statement:**

The specific gap that this project aims to fill is the lack of independent, automated visual evidence from the recipient's side (drop-off location) when the customer is unavailable. Currently, delivery confirmation relies solely on courier-provided photos, which are often inaccurate or fail to verify the package's security within the recipient's storage.

- **Scope & Limitations:**

To ensure this project is completed on time and remains focused, the system limitations are defined as follows:

- **Scope:** **Retrofits** a commercial galvanized steel box with an **external IoT control module (ESP32-S3)** to implement Multi-Factor Authentication (GPS Geofencing < 20m & dynamic OTP via 3x4 Matrix Keypad), solenoid lock control, packages detection (ultrasonic sensor and magnetic reed switches), capture visual evidence inside the container using an OV2640 camera module synchronized with the door mechanism and internal LED flash, also near real-time notification and device management (connectivity status + reception logs) via the mobile application.
- **Limitations:**
 - **Excludes:** Courier face verification; battery and or solar operation (AC power via adapter only); payment system e.g., Cash On Delivery); and direct integration with major e-commerce APIs.
 - **Operational:** Requires active 2.4GHz Wi-Fi for authentication (no offline fallback). Does not feature non-volatile local storage (SD Card) for offline data retention, image buffering relies solely on PSRAM for immediate cloud transmission. The external PLA enclosure provides weather/tamper resistance for electronics but offers lower physical security than the main steel storage.
 - **Environment:** Designed for semi-outdoor (shaded) installation; internal image clarity relies solely on the provided auxiliary flash; soft package

detection (e.g., polymailer packed) may vary due to ultrasonic absorption.

- **Objectives:**

The purpose of developing this system is:

- Achieve a **95%** detection rate for rigid packages and **100%** success rate for saving photos to cloud storage (under active internet connection).
- Execute solenoid lock response **within 2 seconds** of a valid PIN entry.
- Ensure that the latency of detection data sent from the device to the user's smartphone is **no more than 5 seconds** under stable network conditions.
- Ensure visual rendering latency (upload initiation to app display) remains **no more than 20 seconds**.

3. Related Work:

- **Idea sources:**

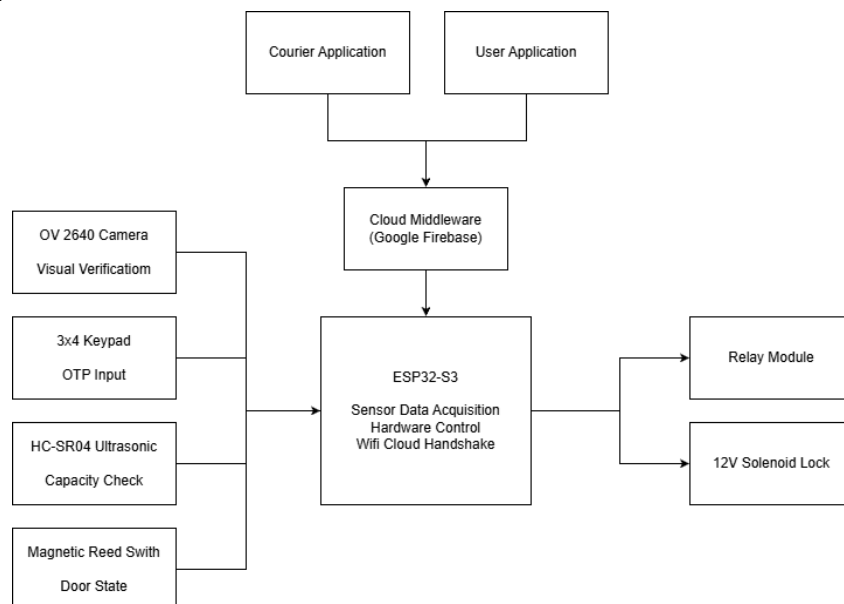
This project was inspired by our own personal experiences. Several occurrences happened where our packages were confirmed as delivered, but in reality, they were sent to the wrong address. A notification would pop up on our phone telling us that the package has arrived and the delivery is finished, but we could not verify if our packages have really arrived in our place or not. This problem could be caused because many couriers have not double checked the address, or because of confusing neighborhood layouts and unclear house numbers that can easily lead to mix-ups. Even current features that require couriers to take a picture for delivery proof, sometimes are not helpful as the pictures themselves are usually not taken clearly. Therefore, we decided to build a smart parcel box that can solve this problem by allowing us to check the package when it arrives.

- **Academic sources:**

https://documentation.espressif.com/esp32-s3_datasheet_en.html

4. System Design and Architecture:

- **Overview:**



- **Hardware needed:**

Microcontroller	ESP32-S3
Sensor (input)	3x4 Matrix Keypad Ultrasonic Sensor (HC-SR04) Magnetic Reed Switch (MC-38) OV2640 Camera Module
Actuator (output)	12V Solenoid Cabinet Lock